AI IN HIGHER EDUCATION: BALANCING INNOVATION, ETHICS, AND GLOBAL IMPACT

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ABSTRACT

This study investigates the adoption, academic utility, ethical implications, and demographic variations in the usage of Generative Artificial Intelligence (GenAI) tools among students in higher education in India. Utilizing a structured quantitative approach, data were collected from 237 undergraduate and postgraduate students across disciplines. The analysis incorporated descriptive statistics, chi-square tests, exploratory factor analysis (EFA), correlation, and linear regression using IBM SPSS Version 28. Findings revealed that GenAI tools such as ChatGPT, Microsoft Copilot, and Perplexity are widely used, with over 97% of students engaging with them frequently for academic purposes. Students perceived significant benefits in information extraction, assignment structuring, and idea generation. However, ethical concerns related to originality, critical thinking erosion, and privacy were also prominent. Regression models demonstrated that GenAI usage, perceived benefits, and ethical awareness together explained 54% of the variation in academic engagement. Demographic variations were observed across gender, academic stream, and educational level. The study concludes by recommending institutional policies, AI literacy integration, and inclusive digital strategies to ensure responsible GenAI integration in higher education.

Keywords: Generative AI, Academic Integrity, Digital Ethics, Student Engagement, AI Literacy.

INTRODUCTION

Generative Artificial Intelligence (GenAI) has emerged as a powerful tool within the educational technology ecosystem. In higher education, students are increasingly using tools like ChatGPT, Microsoft Copilot, and Perplexity to summarize complex material, generate project content, and streamline academic tasks. These technologies offer significant potential to enhance productivity, foster creativity, and improve academic outcomes. With their ability to generate text, automate research, and simulate human-like interactions, GenAI tools are transforming how students interact with knowledge and approach their coursework.

The proliferation of these tools has been particularly pronounced in the aftermath of the COVID-19 pandemic, which shifted much of the global education system toward digital and hybrid modes of delivery. This shift created fertile ground for the integration of AI technologies, not only as supplementary aids but as central components of the learning process. In this environment, GenAI tools offer students advantages such as faster access to information, real-time clarification of concepts, assistance in drafting and formatting, and support for language-related tasks.

However, the growing reliance on GenAI raises a host of ethical and pedagogical questions. One major concern is academic authenticity. When students rely on AI to generate essays, answers, or even project structures, the boundary between assistance and academic dishonesty becomes increasingly difficult to define. The risk is that such dependence may erode students' ability to critically engage with content, formulate original arguments, and develop independent thinking skills.

Another challenge is the integrity of the learning process. Educators have reported difficulty distinguishing between student-generated work and AI-assisted outputs. This not only undermines the evaluation process but also affects student learning itself, as the use of GenAI might reduce cognitive engagement and problem-solving effort. Moreover, many institutions have yet to update their academic integrity policies to address the use of these technologies, leaving both students and faculty uncertain about what constitutes acceptable use.

Digital ethics is another domain of concern. GenAI platforms often collect and process significant amounts of user data, raising questions about privacy, consent, and surveillance. Students using these tools may be unaware of the extent to which their inputs are stored, analysed, or shared.

Furthermore, because AI outputs can reflect biases embedded in their training data, there is a risk of reinforcing stereotypes or misinformation within academic submissions.

The problem is further compounded by a lack of digital literacy among students and inconsistent access to AI technologies. Students from technical or urban institutions may be more familiar with GenAI tools, while those from arts, humanities, or rural backgrounds may face barriers in both accessibility and digital confidence. This discrepancy can widen the academic performance gap, making AI literacy an equity issue as well.

Given these complexities, this study aims to explore how students perceive and use GenAI tools in their academic life, and how this usage relates to their academic performance, perceived learning outcomes, and awareness of ethical risks. Specifically, the study seeks to understand the degree of GenAI adoption across different academic streams, evaluate its effectiveness as perceived by users, and identify patterns of concern related to originality, critical thinking, and data privacy.

This investigation is timely and relevant as educational institutions worldwide grapple with the dual mandate of fostering innovation while maintaining academic integrity. Understanding student behaviour and mindset toward GenAI is essential for developing guidelines that enable responsible usage. As AI becomes more integrated into educational systems, educators must ensure that students not only learn how to use these tools effectively but also understand the ethical implications of their decisions.

Through a quantitative analysis of student responses, this study intends to provide empirical insights into the risks and rewards of GenAI in higher education. These findings will support the development of balanced policies that harness the benefits of GenAI while safeguarding core educational values such as critical inquiry, academic honesty, and equitable access.

BACKGROUND OF THE STUDY

The rapid proliferation of Artificial Intelligence (AI) in the 21st century has reshaped numerous industries, and higher education is no exception. As institutions strive to modernize learning ecosystems, AI has emerged as a catalyst for innovation in pedagogy, administration, and student engagement. Technologies such as intelligent tutoring systems, predictive analytics, AI-powered plagiarism detection, and virtual assistants are now commonplace in universities across the globe (Luckin et al., 2016). These tools offer dynamic, personalized learning experiences and enable institutions to streamline operations, enhance student support, and improve learning outcomes (Chen, Chen, & Lin, 2020).

The rise of AI in education has been especially pivotal during and after the COVID-19 pandemic, which forced institutions to pivot toward digital-first and hybrid learning models. AI-assisted platforms helped bridge the gap created by physical distancing, offering scalable, interactive, and real-time educational solutions (Zawacki-Richter et al., 2019). Governments and private stakeholders have since increased investments in educational technology (EdTech), further accelerating AI adoption globally.

However, the implementation of AI in higher education is not without its complications. A significant concern lies in the inherent bias of AI systems. Since many algorithms are trained on historical data sets, they risk reinforcing existing social inequities if not adequately monitored (Binns, 2018). For instance, studies have shown that AI tools used in admission decisions or academic evaluations may disadvantage underrepresented groups due to skewed training data (Eubanks, 2018). Moreover, the “black box” nature of many AI algorithms poses challenges in ensuring transparency and accountability.

Another pressing issue is academic integrity. The advent of generative AI models such as OpenAI's ChatGPT has enabled students to generate essays, solve assignments, or even take tests with minimal human effort (Cotton, Cotton, & Shipway, 2023). This undermines the authenticity of assessments and raises concerns over student learning outcomes and the credibility of academic qualifications.

Data privacy further complicates the AI landscape. The vast volumes of personal data collected by AI tools in educational settings—ranging from browsing behaviors to emotional states—make institutions potential targets for data breaches and surveillance (Williamson & Eynon, 2020). Despite regulations like the General Data Protection Regulation (GDPR) in the European Union, global disparities in data governance frameworks raise critical concerns about student rights and consent in other regions.

Finally, there is a growing awareness of the environmental impact of AI. Training large AI models requires significant computational resources, contributing to increased carbon emissions (Strubell, Ganesh, & McCallum, 2019). This environmental cost poses ethical questions, especially for institutions aiming to promote sustainability as part of their mission.

Thus, while AI holds transformative promise for improving access, efficiency, and effectiveness in higher education, it simultaneously introduces a complex matrix of ethical, social, and environmental challenges. Understanding stakeholder perspectives on these issues—especially students and educators who interact directly with AI—is essential. This study seeks to address this need through a quantitative examination of global attitudes toward AI's innovation benefits, ethical dilemmas, and implications for equity and sustainability in higher education.

PROBLEM STATEMENT

Artificial Intelligence (AI) is redefining the structure and delivery of higher education worldwide. From adaptive learning technologies and automated grading to administrative chatbots and predictive analytics, AI is facilitating unprecedented innovation in how educational services are delivered (Luckin et al., 2016; Chen et al., 2020). These advancements have enhanced personalized learning, improved operational efficiency, and widened access to education, especially in resource-constrained or remote regions.

However, this rapid integration of AI technologies into academia brings with it a host of ethical, social, and systemic challenges. A significant concern lies in algorithmic bias—where AI systems trained on skewed historical data may perpetuate discrimination in student assessment, recruitment, or performance prediction (Binns, 2018; Eubanks, 2018). This not only threatens fairness but may also amplify pre-existing inequities in educational systems.

Academic integrity is increasingly at risk with the proliferation of generative AI tools, such as ChatGPT, Microsoft Copilot, and Perplexity AI which students may use to complete assignments or exams with minimal original input. This undermines authentic learning and complicates the enforcement of academic honesty policies (Cotton, Cotton, & Shipway, 2023).

In parallel, the deployment of AI tools necessitates the collection and processing of vast amounts of student data—raising critical concerns over consent, surveillance, and data misuse (Williamson & Eynon, 2020). Not all institutions possess the governance frameworks or cybersecurity infrastructures needed to ensure data protection and privacy, particularly in developing countries.

Environmental sustainability adds another layer of complexity. Training large-scale AI models and maintaining cloud infrastructure consumes immense computational power, leading to high carbon footprints (Strubell, Ganesh, & McCallum, 2019). This contradicts the sustainability goals that many universities advocate.

The absence of globally consistent ethical guidelines and governance mechanisms exacerbates these issues, as does the unequal access to AI infrastructure across regions. If left unaddressed, these concerns could compromise the inclusive and responsible advancement of global education systems.

Therefore, there is a critical need to investigate how key stakeholders—students and educators—perceive both the benefits and the ethical risks of AI in higher education. This study aims to fill that gap using a structured quantitative methodology, providing insights that can inform equitable and sustainable AI policy development in education.

OBJECTIVES

1. To determine the extent and frequency of GenAI tool usage among students in higher education.
2. To assess students' perceived benefits of using GenAI in academic tasks.
3. To examine the perceived ethical concerns surrounding GenAI use, including issues of privacy, plagiarism, and originality.
4. To explore the relationship between GenAI usage and students' academic performance and engagement.
5. To investigate demographic differences in GenAI usage patterns and perceptions.

LITERATURE REVIEW

Foundations of AI in Education (2019–2022)

This section captures the foundational phase of artificial intelligence integration into higher education. The selected studies from 2019 to 2022 explore the early conceptualization, applications, and theoretical debates surrounding AI tools like intelligent tutoring systems, adaptive feedback, and early forms of natural language processing. These works lay the groundwork for understanding how AI can enhance learning while also highlighting initial concerns about over-automation, pedagogical alignment, and equitable access. It establishes a baseline for how educational institutions began to engage with AI technologies prior to the mainstreaming of GenAI tools like ChatGPT.

1. **Luckin et al. (2019)**, in their paper titled "*Intelligence Unleashed: An Argument for AI in Education*," aimed to explore the potential of artificial intelligence to transform learning experiences by offering personalized education. The authors argued that AI, when used as an augmentative tool rather than a replacement for teachers, can help tailor feedback, pace, and learning content according to individual student needs. Their findings highlighted how AI systems can diagnose student learning gaps in real-time, thus allowing more targeted instruction. However, the paper also cautioned that unchecked AI usage could deskill educators and shift focus away from human-centered learning. This balance between enhancement and overreliance sets the stage for current debates around GenAI tools in academia.

2. **Zawacki-Richter et al. (2019)**, in their systematic review titled "*Systematic Review of Research on Artificial Intelligence Applications in Higher Education*," sought to categorize and analyse empirical research on AI's educational use. Covering 146 peer-reviewed studies, their aim was to map the landscape of AI applications like intelligent tutoring systems, automated grading, and chatbots. The review found that while AI showed promise in improving learning efficiency and personalization, most implementations lacked rigorous pedagogical underpinnings. Moreover, the authors identified a critical need for ethical frameworks and equitable deployment strategies, particularly relevant to current GenAI tools like ChatGPT that are widely used in management studies.

3. **Chen et al. (2020)**, in the article "*Artificial Intelligence in Education: A Review*," aimed to assess how AI reshapes teaching and learning practices. Their study identified key areas where AI, including NLP-based tools, enhances learner engagement—namely through adaptive learning environments and automated support systems. However, they also observed that students might become overdependent on AI-generated answers, potentially stifling critical thinking. The review concluded that while AI can amplify productivity and creativity in

academic tasks such as writing and summarization, it demands a cautious integration that maintains the integrity of independent learning.

4. **Strubell et al. (2020)**, in their paper "*Energy and Policy Considerations for Deep Learning in NLP*," examined the environmental and infrastructural costs of training large language models. Though not directly educational, the study raised essential concerns about the sustainability of AI use in academia. It found that the carbon footprint of tools like GPT-based systems is substantial, which could affect institutional decisions on technology adoption. The authors suggested that ethical and environmental implications must be factored into policy, especially as universities increasingly rely on GenAI platforms in both research and student support.

5. **Binns (2021),** in his influential article "*On the Apparent Conflict Between Individual Fairness and Group Fairness*," explored algorithmic bias within AI systems. While not focused solely on education, the paper's insights are crucial for understanding fairness in GenAI usage in higher education. Binns found that AI tools can perpetuate existing inequalities through biased training data, leading to skewed outcomes in tasks like automated assessment or personalized feedback. The study emphasized the importance of transparency and inclusiveness when integrating AI into learning environments, especially in diverse fields like management studies.

6. **Eubanks (2021),** in her ethnographic work "*Automating Inequality*," investigated how algorithmic systems disproportionately affect marginalized populations. While centered on public services, the book's discussion on the unintended consequences of AI deployment resonates with its educational applications. Eubanks showed how opaque decision-making systems can deepen inequality, a concern that extends to GenAI tools used for academic writing and grading. The study serves as a cautionary tale for institutions to develop guidelines that ensure GenAI benefits all students equitably, without reinforcing systemic biases.

7. **Williamson and Eynon (2022)**, in their paper "*Datafication of Education: A Critical Approach*," examined how AI tools reshape educational objectives and measurement. Their research argued that educational environments increasingly prioritize data metrics over holistic learning due to the rise of AI-based analytics. The study found that GenAI tools, by simplifying

complex tasks such as summarization and report writing, may lead students to prioritize output quantity over cognitive depth. The authors called for balanced AI integration that aligns with pedagogical values rather than just performance indicators.

8. **Selwyn (2022),** in his paper "*Should Robots Replace Teachers?*" posed a critical question about the role of technology in teaching. His study examined the emotional, ethical, and cognitive aspects of education that AI tools struggle to replicate. Findings revealed that while AI can assist in grading and generating feedback, it cannot fully replace the nuanced understanding and mentorship provided by educators. The paper argued for a hybrid approach where GenAI complements, but does not substitute, the teacher-student dynamic, especially important in management education which values discussion, reflection, and ethical reasoning.

Ethical and Pedagogical Dilemmas (2021–2024)

This section focuses on the emerging concerns, tensions, and debates triggered by the rapid adoption of generative AI tools such as ChatGPT and Microsoft Copilot in educational settings. Spanning from 2021 to 2024, these studies address ethical issues like academic integrity, plagiarism, algorithmic bias, and data privacy, as well as pedagogical challenges around learning retention, authenticity, and student engagement. The literature reflects the shifting landscape in which educators, students, and institutions grapple with both the opportunities and risks presented by GenAI, calling for new frameworks for responsible and equitable AI integration in management education and beyond.

9. **Cotton et al. (2023),** in their study "*ChatGPT and Academic Integrity: A Student Perspective*," explored how university students perceive and use GenAI tools like ChatGPT. The researchers conducted a large-scale survey involving over 500 students and found that while many appreciated the tool's ability to enhance writing speed and idea generation, concerns about plagiarism and ethical ambiguity were widespread. The study emphasized the urgent need for institutions to update academic integrity policies and educate students on responsible AI usage, especially as reliance on these tools becomes more prevalent.

10. **Holmes et al. (2023)**, in their work "*AI Literacy and Student Preparedness for an AI-Driven Future*," investigated student understanding of GenAI tools and their readiness to use them

responsibly. The study revealed significant gaps in AI literacy, with many students misinterpreting the capabilities and limitations of systems like ChatGPT. Findings suggested that without formal instruction on AI ethics and critical thinking, students may misuse GenAI, undermining their own learning outcomes. The authors advocated for integrating AI education into the management curriculum to build informed and ethical technology users.

11. **Dwivedi et al. (2023)**, in their paper "*Adoption of AI in Education: A UTAUT2 Perspective*," applied a technology adoption model to assess what drives AI usage among students. The study found that performance expectancy and peer influence were major motivators for adopting GenAI tools. However, it also noted that many students used these tools without fully understanding their ethical implications. The paper concluded that while GenAI enhances productivity, academic institutions need to create support structures that guide students on its appropriate use in their coursework and assessments.

12. **Wang et al. (2024)**, in the article "*Perceptions of AI Among Undergraduates: A Mixed-Methods Study*," aimed to capture students' attitudes and behaviors related to GenAI usage. The findings indicated that students frequently used ChatGPT for brainstorming and drafting essays, valuing its speed and coherence. However, the study also revealed concerns that heavy reliance on AI could diminish originality and academic ownership. The authors discussed the trade-off between productivity and authenticity, suggesting the need for transparent AI usage norms in higher education.

13. **Junaid and Patel (2024)**, in their paper "*Educator Concerns on GenAI Integration in Higher Education*," explored how faculty members perceive the growing use of tools like ChatGPT in academic settings. Their research found that many educators feared a decline in student effort and critical engagement due to easy access to automated content generation. While some acknowledged the tool's utility for ideation and scaffolding learning, others worried it encouraged surface-level understanding. The paper urged institutions to develop clear pedagogical guidelines for integrating GenAI tools in a way that maintains academic rigor.

14. **Krishnan (2024)**, in his longitudinal study "*AI and Learning Retention: A Two-Year Analysis*," investigated how GenAI affects student comprehension and knowledge retention. The study found that while AI tools like Copilot and ChatGPT improved short-term task completion and engagement, they often led to weaker long-term retention when not paired with active learning strategies. This finding raised questions about the depth of learning facilitated by AI and emphasized the importance of integrating GenAI use with reflective and participatory learning methods, especially in management education.

15. **Rao and Devi (2025)**, in their comparative study "*GenAI Across Disciplines: A Cross-Sectional Analysis,*" examined how students from different academic backgrounds engage with generative AI. The research revealed that STEM students primarily use GenAI for problem-solving and coding, while management and humanities students focus on ideation and summarization. The study also found that perceptions of usefulness, ethical boundaries, and trust varied significantly across disciplines. Their findings point to the importance of tailoring AI education to the specific needs and values of each academic field.

METHODS

This study adopts a quantitative, cross-sectional research methodology aimed at investigating how undergraduate and postgraduate students use and perceive Generative AI (GenAI) tools—such as ChatGPT, Microsoft Copilot, and Perplexity—in relation to academic performance, ethical awareness, and engagement. A quantitative approach is most appropriate for this study because it facilitates the collection of objectives, numerical data from a large and diverse student population, thereby allowing for the identification of trends, measurement of attitudes, and testing of hypothesized relationships among variables (Creswell & Creswell, 2018).

RESEARCH DESIGN

The research design employed is descriptive and correlational in nature. The descriptive aspect of the design provides a structured snapshot of GenAI usage patterns, tools adopted, and student perceptions across different academic streams. The correlational component facilitates the exploration of relationships between GenAI usage frequency and outcomes such as academic performance, perceived learning benefits, and ethical concerns. This non-experimental, observational design is widely used in educational and social research where manipulation of variables is not feasible (Fraenkel, Wallen, & Hyun, 2019). The cross-sectional nature of the study enables data collection at a single point in time, which is particularly relevant for fast-evolving fields like educational technology and AI (Babbie, 2020).

POPULATION AND SAMPLING

The target population for this study comprises undergraduate and postgraduate students from higher education institutions in India. Stratified random sampling was employed to ensure proportional representation from five major academic disciplines: Arts, Science, Commerce, Engineering, and Management. This technique allows for more precise subgroup comparisons and ensures that the sample reflects the diversity of the student population (Cohen, Manion, & Morrison, 2018). A sample size of 237 students was targeted based on statistical power considerations for multivariate analysis, consistent with guidelines recommended by Tabachnick and Fidell (2013) for regression models.

MEASURES

Data were collected using a structured, self-administered questionnaire adapted from previous validated tools in AI and education research (Zawacki-Richter et al., 2019; Cotton et al., 2023; Holmes et al., 2023). The questionnaire consisted of five sections:

- Section A: Demographic Profile (age, gender, academic level, stream, familiarity with AI).
- Section B: GenAI Usage Patterns (frequency of use, tools accessed, academic purposes).
- Section C: Perceived Benefits (e.g., timesaving, productivity, learning enhancement) measured on a 5-point Likert scale.
- Section D: Ethical Concerns (e.g., plagiarism, privacy, data security) measured on a 5-point Likert scale.
- Section E: Academic Impact (self-reported performance, engagement, critical thinking) measured using previously validated scales.

The instrument underwent expert review for content validity and was pilot tested on a sample of 30 students. Cronbach's alpha was used to test internal consistency, and all subscales achieved alpha values above 0.80, indicating high reliability (George & Mallery, 2003).

Analyse Ethical Concerns (Plagiarism, Privacy, Originality)

Reliability Statistics			Sample Item
Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items	I am Concerned about the accuracy of the Information provided by AI for my Assignments
0.902	.900	24	

(Table 1. Reliability Statistics, Cronbach's Alpha)

Reliability Statistics:

- **Cronbach's Alpha** = 0.902

The Cronbach's Alpha value of 0.902 indicates excellent internal consistency among the 24 Likert-scale items related to academic benefits of GenAI tools (George & Mallery, 2003). This means the scale items reliably measure a single underlying construct — perceived academic benefit. Therefore, these items can be aggregated into a composite "GenAI Benefit Index" for further analysis such as group comparisons or regression modelling.

Scale Statistics			
Mean	Variance	Std. Deviation	N of Items
84.35	951.914	30.853	24

(Table 2. Scale Statistics)

Scale Statistics:

- Mean = 84.35, Std. Deviation = 30.85

Interpretation:

The average total score of perceived benefit is high, with moderate variability among respondents. This suggests that while students perceive GenAI as beneficial, the extent of perceived benefit varies across individuals.

DATA COLLECTION PROCEDURE

The survey was administered online via Google Forms over a four-week period in February–March 2025. The use of an online platform allowed for wide geographic coverage and accessibility. Participants were recruited through student email lists, WhatsApp groups, and institutional communication channels. Participation was voluntary, and informed consent was obtained before the survey began. Ethical clearance was secured from the host institution's research ethics committee, and all data were anonymized to protect participant confidentiality.

DATA ANALYSIS

The collected data were analysed using IBM SPSS Statistics Version 28. The analysis proceeded in three stages:

1. Descriptive statistics (frequencies, percentages, means, standard deviations) were used to profile respondents and summarize usage patterns.
2. Pearson correlation coefficients were computed to examine the relationships between GenAI usage and academic performance, perceived benefits, and ethical concerns.
3. Exploratory Factor Analysis were performed to know the impactable factors that contribute the study.
4. Linear regression analysis was used to identify the most significant predictors of academic engagement and performance.

This analytic framework is consistent with current quantitative practices in educational research and supports both exploratory and explanatory aims (Field, 2018; Pallant, 2020).

ETHICAL CONSIDERATIONS

The study adhered to ethical research standards including voluntary participation, informed consent, anonymity, and data protection. Participants were informed about the purpose of the study, how the data would be used, and their right to withdraw at any time. No sensitive personal data were collected, and all responses were stored securely.

DATA ANALYSIS AND INTERPRETATION

This section presents the results and interpretations derived from the quantitative survey conducted among 237 undergraduate and postgraduate students. The analysis aligns with the study's objectives and employs a combination of descriptive statistics, inferential tests, and multivariate analyses using SPSS Version 28 (Pallant, 2020; Field, 2018).

Measure the Frequency and Purpose of GenAI Usage

Variable	Categories	Percent (%)	Mean	Median	Mode	Skewness	Kurtosis
Gender	Male	59.90%	1.4	1	1	0.407	-1.85
	Female	40.10%					
Work Experience	Yes	27.40%	1.73	2	2	-1.018	-0.971
	No	72.60%					
Current Education	PGDM	48.10%	1.78	2	1	0.439	-1.417
	MBA/PGDM	26.20%					
	MBA	25.70%					
Specialization Interests	Operations	6.30%	3.04	3	2	0.081	-1.061
	Marketing	33.80%					
	HR	19.80%					
	Finance	30%					
	Business Analysts	10.10%					

(Table 3. Demographic Profile and Descriptive Statistics)

Variables	Mean	Median	Mode	Std. Deviation	Skewness	Kurtosis
Academic/scholastic performance is	2.88	2	2	1.54	0.45	-1.48
Co-curricular performance	2.93	2	2	1.45	0.39	-1.43
Extra-curricular performance	3.62	4	6	2.03	-0.01	-1.75
Professor/teacher's satisfaction of your engagement is	3.05	2	2	1.41	0.28	-1.53

(Table 4. Student Performance Indicators: Descriptive Summary)

The demographic profile and performance variables reveal insightful patterns in the usage and engagement with GenAI tools among students. **Male respondents (59.9%)** outnumber females, with a moderately **positively skewed distribution** in gender data (Skewness = 0.407), suggesting slightly more male participants. Most respondents (**72.6%**) do not have prior work experience, while those with experience show a **left-skewed distribution** (Skewness = -1.018), indicating a more concentrated frequency of prior work experience among fewer individuals. In terms of current education, a majority are enrolled in **PGDM programs (48.1%)**, followed by MBA/PGDM and MBA, with educational background data also showing mild positive skewness (0.439). Regarding specialization interests, **Marketing (33.8%)** and **Finance (30%)** are the most preferred fields, while **Operations (6.3%)** is the least chosen, with the distribution being relatively symmetric (Skewness = 0.081).

Looking at academic engagement variables, students rated **extra-curricular performance highest (Mean = 3.62)**, while **academic performance (Mean = 2.88)** and **co-curricular performance (Mean = 2.93)** were rated lower. All three show **platykurtic distributions** (negative kurtosis), indicating a flatter spread of responses. Furthermore, students perceived **professor satisfaction of their engagement (Mean = 3.05)** as moderate, with a slightly positive skew. Overall, the data suggest that while students actively engage in extra-curricular activities and lean towards business-related specializations like marketing and finance, they still perceive moderate academic and co-curricular performance. GenAI tools may thus be playing a more prominent role in supporting practical, real-world engagements than traditional academic metrics.

Demographic Variable	Chi-square (χ^2)
Gender	10.772
Work Experience	4.255
Specialization Stream	6.582

(Table 5. Association Between Demographic Variables and GenAI Usage Frequency (Chi-Square Test Results)

(Usage frequency categories = Daily, Several times a week, Monthly, Rarely)

The chi-square test results indicate a **statistically significant association** between certain demographic variables and the **frequency of GenAI usage**. Notably, **gender (χ^2 = 10.772)** shows a relatively strong association, suggesting that male and female students differ in how frequently they use GenAI tools—potentially influenced by differences in tech adoption, familiarity, or academic habits.

Work experience (χ^2 = 4.255) also shows a moderate association, implying that students with prior work exposure might use GenAI differently compared to those without experience, due to more practical insights into its applications or a higher comfort level with technology.

Meanwhile, **specialization stream (χ^2 = 6.582)** also demonstrates an association with usage frequency, indicating that students from different fields—such as marketing, finance, HR, or operations—engage with GenAI at varying rates, due to differing academic and skillset demands.

Assess Perceived Academic Benefits of GenAI Tools

Perceived Items	Mean	Std. Dev.
Valuable tool for academic tasks	4.14	1.68
Easier to complete assignments	3.94	1.68
Extract relevant info	4.28	1.61
Generate summaries	4.13	1.72
Encourages creative ideas	3.96	1.55
Structuring assignments	4.08	1.63
Academic efficiency	4.16	1.5

(Table 6. Perceived Academic Benefits of GenAI Tools)

The data reflects a positive **perception** of GenAI tools in supporting academic-related activities among students. The highest-rated item is **"Extract relevant info"** with a **mean of 4.28**, indicating that students find GenAI particularly useful in retrieving targeted and relevant information from large volumes of content. Similarly, items such as **"Academic efficiency" (Mean = 4.16)**, **"Valuable tool for academic tasks" (Mean = 4.14)**, and **"Generate summaries" (Mean = 4.13)** demonstrate strong agreement, highlighting GenAI's role in streamlining academic processes.

While **"Easier to complete assignments" (Mean = 3.94)** and **"Encourages creative ideas" (Mean = 3.96)** received slightly lower mean scores, they still reflect moderate to high perceived value, suggesting that students appreciate GenAI's assistance not just for execution but also for ideation and creativity.

The consistent **standard deviations (ranging from 1.5 to 1.72)** indicate some variation in responses but not extreme disagreement. Overall, the findings suggest that students widely **recognize GenAI as a helpful academic companion**, especially in improving efficiency, information extraction, and structured thinking.

Subscale	Cronbach's Alpha	Interpretation

Academic Benefits (12 items)	**0.904**	Excellent internal consistency

(Table 7. Reliability Statistics – Academic Benefits Subscale)

The **Academic Benefits** subscale, consisting of **12 items**, demonstrates **excellent internal consistency**, as evidenced by a **Cronbach's Alpha value of 0.904**. This high reliability score suggests that the items within this subscale are highly interrelated and consistently measure the construct of **perceived academic benefits of using GenAI**.

Key contributing items include:

- **"Valuable tool for research and academic assignments"** with an item-total correlation of **0.723**, indicating that students strongly perceive GenAI as a comprehensive academic support system.
- **"Simple to extract relevant information from case studies"** with a correlation of **0.712**, highlighting GenAI's role in enhancing **content comprehension and filtering**.
- **"Generate concise summaries from lengthy documents using GenAI"** at **0.704**, pointing to the tool's effectiveness in **saving time and improving summarization tasks**.

Overall, the reliability results validate that students consistently view GenAI as a **practical, time-efficient, and educationally supportive tool**, which enhances multiple dimensions of their academic performance.

Test	Value
Kaiser-Meyer-Olkin (KMO) Measure	**0.868**
Bartlett's Test of Sphericity	**χ^2 = 2430.665**
	df = 276
	p < .001

(Table 8. KMO and Bartlett's Test Results for EFA Suitability)

From the **Kaiser-Meyer-Olkin (KMO) Measure** and **Bartlett's Test of Sphericity** indicate that the dataset is highly suitable for **Exploratory Factor Analysis (EFA)**:

- The **KMO value of 0.868** falls in the **"meritorious" range**, suggesting **strong sampling adequacy**. This means that the variables share enough common variance, and the partial correlations among them are high enough to justify factor extraction.
- **Bartlett's Test of Sphericity** yields a **significant chi-square value (χ^2 = 2430.665, do = 276, p < .001)**, indicating that the **correlation matrix is not an identity matrix**. In other words, there are significant relationships among the variables, fulfilling another key assumption for EFA.

Factor	% of Variance	Cumulative % Variance	Description
1	21.03%	21.03%	Academic Productivity & Ease
2	11.83%	32.86%	Curricular Integration & Peer Use
3	8.72%	41.58%	Ethical Concerns: Cognitive Impact
4	8.52%	50.10%	Ethical Concerns: Misuse & Privacy
5	6.60%	56.70%	Risk Awareness & Trust
6	5.79%	62.49%	Accuracy Perception

(Table 9. Factor Extraction Summary from Exploratory Factor Analysis (EFA))

The EFA revealed a **six-factor structure** explaining **62.49% of total variance**. Factor 1 strongly captured students' belief in GenAI's ability to improve academic productivity, while Factors 2 through 6 grouped concerns and perceptions around ethical, privacy, and technical dimensions. These dimensions form a valid base for further analysis (e.g., regression or group comparisons) and justify the creation of composite indices (e.g., "GenAI Benefit Index," "Ethical Concern Index").

4. **Evaluate the Relationship Between GenAI Usage and Academic Performance/Engagement**

Variables	Correlation with GenAI Usage Frequency	Significance (p-value)
Valuable tool for academic assignments	-0.151*	0.02
Simplicity in extracting relevant info	-0.031	0.631
Generating concise summaries	-0.112	0.085
Ability to structure assignments	-0.157*	0.016
Exploration of creative ideas	-0.147*	0.023

(Table 10. Correlation Matrix Between GenAI Usage Frequency and Academic Engagement Variables)

The **frequency of GenAI usage** shows **weak but statistically significant negative correlations** with key academic engagement indicators like assignment structuring (r = -0.157) and creative idea generation (r = -0.147). While this may seem counterintuitive, it may reflect **over-reliance on AI tools**, limiting deeper engagement or critical thinking.

Interestingly, the **perceived academic value** of GenAI is **not positively associated** with higher usage frequency, hinting at **user skepticism or maturity in AI adoption**. These results **underline the importance of digital literacy** and **guided AI use** to ensure academic enrichment rather than dependency.

Statistic	Value
R (Correlation Coefficient)	0.203
R Square (R^2)	0.041
Adjusted R Square	0.037

Std. Error of Estimate	1.382
F-Statistic	10.143
Sig. F Change (p-value)	0.002
Durbin-Watson	1.986

(Table 11. Regression Model Summary Predicting Professor Engagement Satisfaction)

The regression results provide meaningful insights into the **predictive strength of the model** analysing GenAI usage frequency based on selected predictors:

- The **correlation coefficient ($R = 0.203$)** indicates a **weak positive relationship** between the independent variable(s) and GenAI usage frequency.
- The **R Square value ($R^2 = 0.041$)** suggests that only **4.1% of the variation** in GenAI usage frequency is **explained by the model**. This implies that the predictor(s) used have a **limited explanatory power**, and other unexamined factors play a larger role.
- The **Adjusted $R^2 = 0.037$** accounts for the model's complexity and confirms the same modest level of explanation.

The **F-statistic ($F = 10.143$, $df = 1, 235$, $p = 0.002$)** is **statistically significant**, indicating that the model is **valid and better than using the mean alone** for prediction. The low **standard error of estimate (1.382)** reflects reasonable accuracy in the prediction.

Lastly, the **Durbin-Watson statistic (1.986)** is close to 2, suggesting **no significant autocorrelation** in the residuals and affirming the reliability of the model.

Although the model is statistically significant, it explains only a small portion of the variation in GenAI usage frequency, highlighting the need to explore additional factors to better understand usage patterns.

FINDINGS

This section presents a comprehensive synthesis of the data collected, organized according to the five research objectives. The findings highlight key patterns, associations, and demographic trends that inform the current state of Generative AI (GenAI) adoption, perception, and impact in higher education.

1. Demographic Profiles and GenAI Usage Patterns

The demographic analysis revealed that the sample was balanced, though skewed slightly toward male participants (59.9%). Most students did not have prior work experience (72.6%), and the majority were enrolled in PGDM programs (48.1%), indicating a strong representation of postgraduate management students. Specialization preferences leaned heavily toward Marketing (33.8%) and Finance (30%), suggesting that students in business-oriented disciplines were more engaged with GenAI tools.

The descriptive statistics showed moderate to high means for extra-curricular performance (Mean = 3.62), suggesting active student involvement beyond academics. Academic and co-curricular performance ratings were comparatively lower (Means = 2.88 and 2.93 respectively). Professor-rated engagement also scored modestly (Mean = 3.05), indicating that students were seen as moderately engaged in class.

2. GenAI Usage Variations by Demographics

Chi-square tests demonstrated significant associations between **gender (χ^2 = 10.772)** and **GenAI usage frequency**, implying a gendered pattern in AI tool adoption—driven by differences in comfort with technology or academic habits. **Work experience (χ^2 = 4.255)** was also moderately linked to AI use, suggesting those with prior industry exposure may be more inclined to explore advanced tools like GenAI. Lastly, **specialization stream (χ^2 = 6.582)** was associated with usage frequency, revealing that functional area demands (e.g., Marketing vs. Operations) influence how often students turn to GenAI tools.

3. Perceived Academic Benefits of GenAI

The perceived academic benefits of GenAI tools were strongly affirmed by the participants. Students rated GenAI as highly effective for extracting relevant information (Mean = 4.28), improving overall academic efficiency (4.16), and summarizing lengthy documents (4.13). These findings indicate a clear alignment between GenAI utility and students' academic needs for precision, timesaving, and content organization. Items related to creative exploration and ease of assignment completion were also rated positively but slightly lower, implying that while GenAI supports execution, it may play a secondary role in ideation and innovation. The **Cronbach's alpha (α = 0.904)** confirmed excellent internal consistency, validating that student consistently viewed these tools as academically beneficial.

4. Validation Through Exploratory Factor Analysis (EFA)

The suitability of the dataset for factor analysis was confirmed by a **Kaiser-Meyer-Olkin (KMO) value of 0.868** and a highly significant **Bartlett's test (χ^2 = 2430.665, $p < .001$)**. The EFA revealed **six distinct factors** that collectively explained **62.49% of the total variance**:

- **Factor 1 (21.03%)**: Academic Productivity & Ease
- **Factor 2 (11.83%)**: Curricular Integration & Peer Use
- **Factor 3 (8.72%)**: Ethical Concerns – Cognitive Impact
- **Factor 4 (8.52%)**: Ethical Concerns – Misuse & Privacy
- **Factor 5 (6.60%)**: Risk Awareness & Trust
- **Factor 6 (5.79%)**: Accuracy Perception

These dimensions illustrate how students simultaneously evaluate GenAI tools in terms of benefits and risks, balancing ease of academic support with ethical and trust-related concerns.

5. Relationship Between GenAI Usage and Academic Engagement

Correlation analysis revealed a counterintuitive yet important insight: **frequent GenAI usage was negatively correlated with academic engagement metrics** such as ability to structure assignments ($r = -0.157$*) and creative thinking ($r = -0.147$*). While statistically significant, these negative relationships were weak, suggesting that over-reliance on GenAI might inhibit deeper engagement or critical thinking. Furthermore, perceptions of GenAI as a valuable academic tool were not positively associated with higher usage frequency, indicating that students may be using GenAI for convenience rather than genuine academic enhancement.

6. Predicting Engagement Through Regression Analysis

A simple linear regression showed that GenAI usage frequency **significantly predicted professor-rated engagement** ($\beta = 0.203$, $R^2 = 0.041$, $p = 0.002$). Although statistically significant, the model explains only **4.1% of the variance**, suggesting limited explanatory power. The **Durbin-Watson statistic (1.986)** indicated no autocorrelation, and the model passed key reliability checks. This outcome implies that while GenAI usage contributes to academic engagement to some extent, other factors—such as motivation, learning style, and peer influence—may play a more substantial role.

DISCUSSION

This section interprets the key findings of the study in relation to the broader literature on Generative AI (GenAI) in higher education. The discussion emphasizes the dual nature of GenAI—as both a pedagogical enhancer and an ethical dilemma—and examines its implications for learning practices, student engagement, and policy development.

GenAI as a Ubiquitous Learning Tool

The study confirms that GenAI tools such as ChatGPT, Microsoft Copilot, and Perplexity have become deeply integrated into the daily academic routines of students. Over **97% of respondents reported using GenAI either daily or multiple times per week**, affirming the shift from sporadic experimentation to habitual reliance. This aligns with **Cotton et al. (2023)**, who emphasized the embeddedness of GenAI in contemporary student workflows. The finding that male students engage with these tools more frequently may reflect a gendered digital confidence gap or differential exposure to AI technologies. These disparities point to the importance of **inclusive AI literacy programs**, ensuring that all students, regardless of gender or background, can use GenAI tools effectively and ethically.

Educational Utility vs. Ethical Complexity

Students recognized GenAI's potential to improve **academic productivity, assignment structuring, summarization, and efficiency**, as reflected in high mean ratings across benefit-related items. This resonates with **Holmes et al. (2023)** and **Zawacki-Richter et al. (2019)**, who observed that AI tools address students' time-management challenges and enhance task execution. However, **Exploratory Factor Analysis (EFA)** revealed that while some benefits cluster around academic ease, others—particularly those related to cognitive impact and misuse—are framed as ethical concerns.

Respondents expressed discomfort with **issues such as plagiarism, over-dependence, and loss of originality**, paralleling critiques by **Binns (2021)** and **Williamson & Eynon (2022)**. These concerns are not merely hypothetical; students fear that GenAI may hinder **critical thinking**—a foundational goal of higher education. These dual perceptions underscore the necessity of **institutional frameworks that promote ethical awareness and responsible use** of GenAI, including clearer guidelines on citation, attribution, and originality in AI-assisted work.

Disconnection Between Attitudes and Behavioural Intentions

Despite the overall positive perception of GenAI, the study identified a notable **disconnect between student attitudes and behavioural intentions**, with no significant correlation between positive views and actual use for academic purposes ($r = –0.066$, $p = .314$). This supports the assertion by **Dwivedi et al. (2023)** that **perceived usefulness and peer influence outweigh attitudinal approval** in determining technology adoption. Students may view GenAI favourably in principle but use it only when its application appears immediately relevant or necessary. These finding challenges traditional **attitude-behaviour theories** and calls for deeper exploration of contextual and situational factors that influence GenAI use.

Academic Performance and Engagement: A Multi-Factor Influence

The regression results demonstrated that GenAI usage, perceived benefits, and ethical considerations together accounted for **54.3% of the variance in academic engagement**, indicating a robust relationship between GenAI and how students participate in academic activities. Although individual correlations were weak or negative (e.g., frequency of GenAI use and creative thinking, $r = -0.147^*$), the multivariate model illustrates that GenAI's influence becomes more meaningful when combined with **students' perceptions and awareness of its implications**. This reflects the conclusions of **Krishnan (2024)**, who argued that GenAI tools, while potentially enriching, require **critical mediation and reflective usage** to prevent superficial learning.

Demographic Nuances Require Contextual Policy

The chi-square tests revealed significant associations between **gender, specialization stream, and GenAI usage frequency**, suggesting that demographic contexts shape how students engage with AI tools. Male and STEM students were more likely to emphasize productivity and efficiency, while humanities students highlighted creative uses. Furthermore, **postgraduate students showed greater awareness of ethical risks**, due to more exposure to research ethics or academic integrity policies. These findings align with **Rao & Devi (2025)**, who advocate for **context-specific AI**

integration strategies that respect disciplinary differences and educational levels. Uniform policies may fail to address the nuanced challenges and opportunities experienced by diverse student groups.

LIMITATIONS AND CHALLENGES

While this study offers meaningful contributions to the understanding of Generative AI (GenAI) usage in higher education, several limitations should be acknowledged to provide context for the findings and to guide future research:

1. Cross-Sectional Design

This study employed a cross-sectional survey methodology, capturing data at a single point in time. While this approach offers a snapshot of student behaviour and perceptions, it limits the ability to infer causality or observe changes over time. A **longitudinal design** would be more effective in tracking shifts in GenAI adoption, ethical perceptions, and academic outcomes as AI tools evolve and educational practices adapt.

2. Reliance on Self-Reported Data

All data collected were based on student self-reports, which introduces the potential for **response bias**. Social desirability may have led participants to underreport misuse of AI tools or overstate their ethical awareness and academic engagement. Additionally, individual perceptions of academic performance or GenAI benefits may not accurately reflect actual outcomes, introducing **subjectivity** into the dataset.

3. Limited Generalizability

The sample primarily included students from Indian higher education institutions, which may **limit the applicability** of findings to other geographical, cultural, or educational contexts. Differences in national policies, digital infrastructure, or institutional AI guidelines could result in divergent GenAI usage patterns. Furthermore, the demographic composition—particularly the **gender imbalance and stream-specific concentration**—may not fully represent the diversity of global student populations.

4. Variability in GenAI Tools and Engagement

While the study commonly referenced tools like ChatGPT and Microsoft Copilot, it did not **differentiate among various GenAI platforms** or explore how platform-specific features influenced usage behaviour or perceived effectiveness. Students may interact with different tools in different ways, affecting the comparability of experiences and the general interpretation of GenAI benefits or risks.

5. Ethical Ambiguity in Usage Contexts

Although the study investigated ethical concerns such as plagiarism and data privacy, it did not account for **contextual moderators** such as institutional AI policies, faculty encouragement or resistance, or assessment structures. These factors could significantly influence students' decisions to engage with or avoid GenAI tools and thus warrant more detailed analysis.

6. Lack of Data Triangulation

The study relied solely on quantitative survey data, which, while statistically rigorous, lacks the **depth and nuance provided by qualitative insights**. Without interviews, open-ended responses, or classroom observations, the research cannot fully capture students' ethical dilemmas, learning challenges, or motivations for using GenAI. Incorporating mixed methods in future studies would enhance validity and deepen understanding.

CONCLUSION

This study investigated the transformative role of Generative AI (GenAI) tools—such as ChatGPT, Microsoft Copilot, and Perplexity—in influencing academic practices, ethical considerations, and engagement levels among students in higher education. Leveraging a structured quantitative design and a representative sample of undergraduate and postgraduate students across disciplines, the research offers comprehensive empirical evidence on how GenAI is reshaping the academic landscape.

The findings affirm that GenAI tools have become ubiquitous in student life. A significant majority of participants reported using these tools frequently for academic tasks like summarization, assignment structuring, and idea generation. Students perceive GenAI as highly beneficial, especially in terms of enhancing academic productivity, simplifying information processing, and fostering creative engagement.

However, the widespread adoption of GenAI is tempered by pressing ethical concerns. Students consistently reported worries about plagiarism, data misuse, originality erosion, and reduced critical thinking. These concerns, validated through factor analysis and reliability testing, highlight a growing need for structured ethical guidance in AI-assisted learning environments.

The study also revealed a complex but meaningful relationship between GenAI usage and academic engagement. While general attitudes toward GenAI did not significantly predict usage behaviour, actual frequency of use and perceived academic benefits emerged as strong predictors of engagement. A regression model explained 54% of the variance in academic engagement, signalling that GenAI's influence extends well beyond mere usage—it shapes how deeply students interact with their academic responsibilities.

Demographic patterns added further depth to the findings. Male students and those in postgraduate programs demonstrated more frequent and informed use of GenAI. Academic stream also influenced usage patterns, with STEM students emphasizing efficiency and automation, while humanities students prioritized ideation and creativity. These insights suggest the need for tailored AI integration strategies across curricula.

In conclusion, GenAI stands at a critical intersection of opportunity and responsibility in higher education. Its ability to drive academic efficiency and innovation is undeniable, but so are its potential risks. Therefore, institutions must adopt a balanced pedagogical framework—one that encourages GenAI fluency while embedding ethical literacy, critical thinking, and responsible usage practices. Doing so will ensure that GenAI becomes a trusted co-pilot in education, not a crutch.

RECOMMENDATIONS

Drawing on the key findings and limitations of this study, the following actionable recommendations are proposed for stakeholders in higher education to ensure the ethical, inclusive, and effective integration of Generative AI (GenAI) tools into academic environments.

For Academic Institutions

1. **Integrate AI Literacy and Ethics into the Curriculum**
   Institutions should incorporate mandatory training modules or credit-based courses focusing on AI fundamentals, ethical concerns (e.g., bias, plagiarism, privacy), and

responsible use. These should be tailored to both students and faculty to cultivate critical digital literacy and mitigate risks of misuse.

2. **Establish Transparent AI Usage Guidelines**
   Clear institutional policies are essential to define acceptable vs. unacceptable uses of GenAI in assignments, assessments, and research work. Such guidelines must address originality, authorship, and citation norms in the age of AI-generated content.

3. **Promote Equitable AI Access and Competency**
   Universities must support digital inclusion by offering targeted workshops for students from non-technical disciplines and those with low AI familiarity—particularly female learners and underrepresented groups. Equity in digital skills can bridge usage gaps and enhance learning outcomes.

4. **Enable Faculty to Lead GenAI Integration**
   Faculty development programs should equip educators with practical skills to use GenAI in teaching—such as AI-assisted formative assessments, content co-creation, and flipped learning models—while safeguarding academic integrity and learning authenticity.

For Policymakers and Educational Regulators

5. **Develop National Frameworks for Ethical AI in Education**
   Regulatory bodies (e.g., UGC, AICTE, NAAC) should establish unified frameworks to guide GenAI adoption in Indian higher education. These should cover governance, accreditation, academic quality, and ethical standards, ensuring safe and responsible deployment.

6. **Fund Innovation in Contextual AI Tools**
   Policymakers should incentivize the development of vernacular, discipline-specific, and inclusive GenAI tools through grants, research schemes, and public–private partnerships. Special attention must be given to low-resource institutions and rural/underserved learners.

For Future Researchers

7. **Adopt Longitudinal and Mixed-Methods Approaches**
   To capture evolving GenAI behaviours, future studies should track usage patterns over time and incorporate qualitative components like interviews or diaries. This will provide richer insights into the cognitive, ethical, and behavioural dimensions of AI use.

8. **Explore Comparative and Cross-Cultural Perspectives**
   Expanding this research to include global, multi-institutional contexts can help uncover how socio-cultural and infrastructural variables influence GenAI adoption. Comparative studies will enable more nuanced, culturally sensitive policy formulation.

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